

AYUSH SINGH TOMAR

AI Agent & RAG Developer — Building Production LLM Systems

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PROFESSIONAL SUMMARY

Final-year Information Technology student specializing in retrieval-augmented generation (RAG) and agentic AI systems. Built and deployed production LLM applications — retrieval pipelines with cited, grounded answers, multi-agent orchestration with LangGraph, LLM fine-tuning (LoRA/PEFT), and Model Context Protocol (MCP) tool serving — each backed by CI-integrated evaluation to catch regressions before deployment. Freelance AI Developer, 5.0/5.0 client rating on Upwork. Looking for a full-time AI Engineer, GenAI Developer, or Machine Learning Engineer role.

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, SQL, HTML, CSS

Generative AI & Agentic Systems: LangGraph, LangChain, CrewAI, Model Context Protocol (MCP), Retrieval-Augmented Generation (RAG), LLM Fine-Tuning (LoRA/PEFT), Prompt Engineering, Groq API, OpenAI API, NLP

ML & Data Science: scikit-learn, PyTorch, Hugging Face Transformers, Random Forest, Model Evaluation

Backend & Frontend: FastAPI, Flask, REST APIs, React, Next.js

Data, Cloud & DevOps: Supabase, PostgreSQL, pgvector, Chroma, SQLite, Vercel, Render, Docker, Git, CI/CD

Core CS: Data Structures & Algorithms (Python), Object-Oriented Programming, System Design fundamentals

PROFESSIONAL EXPERIENCE

Freelance AI Developer — Independent & Client Projects | May 2026 – Present

- Design and deploy production RAG and agentic AI systems end to end — retrieval pipelines, multi-agent orchestration, and LLM tooling — using LangChain, LangGraph, FastAPI, and Groq
- Took a client AI chatbot contract from requirements to delivery in 3 days through Upwork's Talent Accelerator program, closing at a 5.0/5.0 rating; built the conversation flows and wired the Groq API to the client's query logic myself
- Nothing here stays a prototype — every project ships with a working demo link

AI PROJECTS

AskMyDocs — RAG Document Q&A System

[Live Demo](#) • [Source Code](#)

Next.js, Supabase, pgvector, Cohere, Groq API

- Answers document-based questions in under 3 seconds and returns the exact source chunk plus a similarity score behind every answer
- Diagnosed a query/document embedding asymmetry bug — questions were being embedded with the wrong Cohere input type, silently degrading retrieval relevance despite correct chunk embeddings; fixed it and hardened retrieval with hybrid search (RRF)
- Built an LLM-as-judge evaluation suite wired into CI, then found and fixed a scoring bug in the judge itself that was falsely penalizing correct answers — regressions now get caught before a deploy, not after

SalesAgent — Autonomous B2B Sales Agent

[Live Demo](#) • [Source Code](#)

LangGraph, FastAPI, React, scikit-learn, Groq API, Tavily

- Takes a LinkedIn URL, researches the company through a LangGraph node, and scores lead quality with a Random Forest model
- Drafts a personalized cold email referencing real company events, end to end in under 45 seconds
- Built a self-authored eval harness that caught 2 production defects — including uniform lead scores — before they shipped

LoRA Fine-Tuned Resume Screener

[Live Demo](#) • [Source Code](#)

Qwen2.5-0.5B, LoRA/PEFT, PyTorch, Hugging Face Transformers — Published on Hugging Face

- Trained a rank-16 LoRA adapter on just 0.44% of total model parameters so structured resume-fit verdicts are the model's default output, not a prompted-for behavior — validation loss tracked training loss across 3 epochs with no divergence
- Benchmarked fine-tuned vs. zero-shot on the same 96-example eval set: verdict accuracy went from 17.7% to 88.5% and mean score error dropped from 32.03 to 5.39 points — fine-tuning fixed judgment, not just format

AgentLoop — Multi-Step Autonomous Research Agent

[Live Demo](#) • [Source Code](#)

FastAPI, LangGraph, Groq API, Tavily, SQLite, SSE

- Breaks a research question into sub-questions, searches the web, then re-reads its own notes to find gaps and loops back before writing the final report
- Two-tier memory (short-term run state, long-term SQLite recall) streams every reasoning step live to the UI

EDUCATION

B.Tech, Information Technology — Madhav Institute of Technology and Science, Gwalior

2023 – 2027 • CGPA: 7.87 / 10.0

CERTIFICATIONS

Cloud Computing and Distributed Systems — NPTEL (IIT Kanpur), Elite Certification, Top 5%, Score: 90% (Jan–Mar 2026)